

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Describe the **significance of data cleaning** and explain the impact of **poor data quality** on model performance
- Identify and describe various **data cleaning techniques** for handling issues like missing data, duplicates, outliers, and incorrect formatting
- Describe the purpose of **normalisation** and **standardisation** as preprocessing steps
- Describe the role of **feature selection** in improving model efficiency and preventing overfitting
- Describe the three main strategies for feature selection: **filter, wrapper, and embedded** methods

Key Vocabulary

Term	Definition
GIGO (Garbage In, Garbage Out)	The principle that the quality of a model's output depends on the quality of its input data
Data cleaning	The process of detecting and correcting (or removing) corrupt or inaccurate records from a dataset
Missing data	Values that are absent from a dataset; can be handled by deletion or imputation
Imputation	The process of replacing missing values with substituted values (e.g., mean, median)
Outlier	A data point that differs significantly from other observations
Normalisation	Scaling numeric data to a range (e.g., 0 to 1)
Standardisation	Scaling numeric data to have a mean of 0 and a standard deviation of 1
Feature selection	The process of selecting the most relevant features (attributes) for a model
Filter method	Feature selection based on statistical measures, independent of the model
Wrapper method	Feature selection using the model's performance to evaluate feature subsets
Embedded method	Feature selection performed during model training
Overfitting	A model that learns noise in the training data and performs poorly on new data

Approaches to Learning (ATL) Elements

Skill	Description
Thinking (Critical Thinking)	Analysing "dirty" datasets; evaluating trade-offs between different data cleaning techniques
Problem-Solving	Applying a systematic process to identify issues and propose cleaning steps
Communication	Collaborating in pairs; articulating cleaning decisions

2. The Importance of Data Cleaning

The Gourmet Chef Analogy

"Can a world-class chef make a delicious meal with rotten, low-quality ingredients?"



The GIGO Principle

GIGO (Garbage In, Garbage Out): The quality of a model's output depends entirely on the quality of its input data.

Why Data Cleaning Matters

Reason	Explanation
Model Accuracy	Dirty data leads to incorrect predictions
Fairness	Biased data leads to biased models
Reliability	Inconsistent data produces unreliable results
Efficiency	Clean data reduces training time
Interpretability	Clean data makes models easier to understand

Real-World Impact of Poor Data Quality

Industry	Impact
Healthcare	Misdiagnosis due to incorrect patient data
Finance	Incorrect credit risk assessments
Retail	Poor customer recommendations
Autonomous Vehicles	Incorrect object detection leading to accidents

3. Data Cleaning Techniques

Common Data Quality Issues

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COMMON DATA QUALITY ISSUES

MISSING DATA

- Values are absent from the dataset
- Solutions: Deletion or Imputation

DUPLICATES

- Repeated records in the dataset
- Solutions: Remove duplicate rows

OUTLIERS

- Data points that differ significantly
- Solutions: Remove, cap, or transform

INCONSISTENT FORMATS

- Data in different formats (e.g., date formats)
- Solutions: Transform to standard format

INCORRECT VALUES

- Typos, invalid data, impossible values
- Solutions: Correct or filter out

3.1 Handling Missing Data

Method	Description	When to Use
Deletion	Remove rows with missing values	When data is missing at random; when few records are affected
Mean/Median Imputation	Replace missing values with mean or median	For numeric data with a normal distribution
Mode Imputation	Replace missing values with mode (most frequent)	For categorical data
Predictive Imputation	Use a model to predict missing values	When data is not missing at random; advanced technique

Example:

```
python

# Before imputation
Age: [25, 30, None, 40, 35, None]

# Mean imputation
Mean age = (25 + 30 + 40 + 35) / 4 = 32.5
Age: [25, 30, 32.5, 40, 35, 32.5]
```

3.2 Handling Outliers

Method	Description	When to Use
Removal	Delete outlier data points	When outliers are clearly errors
Capping	Limit outliers to a maximum/minimum value	When outliers should be kept but constrained
Transformation	Apply log or square root transformation	To reduce the impact of outliers

Analogy: Outliers are like extreme scores in a class—one student scoring 100% and another scoring 10% might both be unusual.

3.3 Handling Duplicates

Method	Description
Detection	Identify duplicate rows based on key columns
Removal	Keep only one copy of each duplicate record

Analogy: Duplicates are like having multiple copies of the same file—they waste space and can cause confusion.

3.4 Handling Inconsistent Formats

Issue	Solution
Date Formats	Standardise to a single format (e.g., YYYY-MM-DD)

Issue	Solution
Categorical Values	Use consistent values (e.g., "Yes"/"No" not "Y"/"N" or "yes")
Text Case	Convert all text to lowercase or uppercase
Units	Standardise units (e.g., all ages in years)

Example:

```
python

# Before standardisation
Dates: ["01/15/2024", "2024-02-20", "March 5, 2024"]

# After standardisation (YYYY-MM-DD)
Dates: ["2024-01-15", "2024-02-20", "2024-03-05"]
```

4. Normalisation and Standardisation

What is Scaling?

Scaling is the process of adjusting the range of numeric features to ensure they contribute equally to the model.

Normalisation vs. Standardisation

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NORMALISATION VS. STANDARDISATION

NORMALISATION

Scales data to a specific range (usually 0 to 1)

Formula: $(x - \min) / (\max - \min)$

Example: Age [18, 25, 30, 45, 65]
→ [0.0, 0.16, 0.26, 0.52, 1.0]

When to use: When you need bounded values

STANDARDISATION

Scales data to have mean 0 and standard deviation 1

Formula: $(x - \text{mean}) / \text{standard_deviation}$

Example: Age [18, 25, 30, 45, 65]
→ [-1.28, -0.64, -0.24, 0.64, 1.52]

When to use: When you need a normal distribution

Why Scale Data?

Reason	Explanation
Fairness	Features with larger ranges won't dominate the model
Convergence	Training algorithms converge faster on scaled data
Distance-Based Algorithms	Distance-based models (e.g., k-NN) require scaled data

5. Feature Selection

What is Feature Selection?

Feature selection is the process of identifying and retaining the most relevant features (attributes) in a dataset. It removes irrelevant or redundant features.

Why Feature Selection Matters

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WHY FEATURE SELECTION MATTERS

REDUCES OVERFITTING

Too many features can cause the model to learn noise instead of patterns.

IMPROVES EFFICIENCY

Fewer features = faster training and inference

IMPROVES INTERPRETABILITY

Fewer features = easier to understand the model

REDUCES MEMORY USAGE

Less data = less storage and memory required

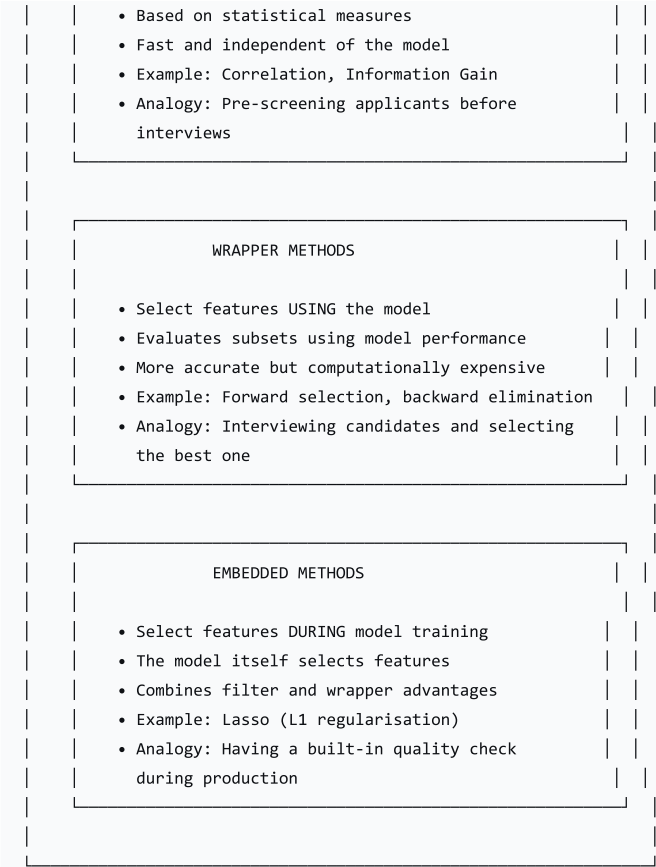
The Three Feature Selection Strategies

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THREE FEATURE SELECTION STRATEGIES

FILTER METHODS

- Select features BEFORE the model



5.1 Filter Methods

Aspect	Description
When	Before model training
How	Statistical measures (correlation, mutual information)
Speed	Fast
Advantage	Independent of the model
Disadvantage	May miss feature interactions

Examples:

Method	Description
Correlation	Remove highly correlated features
Information Gain	Select features with high information gain
Chi-Square Test	For categorical features

Analogy: Pre-screening job applicants based on their CV before interviews.

5.2 Wrapper Methods

Aspect	Description
When	During model training (trial and error)
How	Use model performance to evaluate feature subsets
Speed	Slow (computationally expensive)
Advantage	Finds the best subset for the specific model
Disadvantage	Can overfit; high computational cost

Examples:

Method	Description
Forward Selection	Start with no features; add one at a time
Backward Elimination	Start with all features; remove one at a time

Analogy: Interviewing candidates and selecting the best one based on performance.

5.3 Embedded Methods

Aspect	Description
When	During model training (built-in)
How	The model itself selects features during training
Speed	Medium (faster than wrapper, slower than filter)
Advantage	Combines accuracy of wrapper with speed of filter
Disadvantage	Specific to certain models

Examples:

Method	Description
Lasso (L1 Regularisation)	Penalises features; forces some coefficients to zero
Decision Trees	Trees automatically select features

Analogy: Having a built-in quality check during production.

Comparison Table

Feature	Filter	Wrapper	Embedded
When	Before model	During model (trial)	During model (built-in)
Speed	Fast	Slow	Medium
Model-Dependent	No	Yes	Yes
Risk of Overfitting	Low	High	Medium

Feature	Filter	Wrapper	Embedded
Analogy	Pre-screening	Interviewing	Built-in quality check
Examples	Correlation, Info Gain	Forward Selection	Lasso, Decision Trees

6. Practical Activity: "Clean the Sheet"

Instructions

1. Students work in pairs as "Data Janitors"
2. Examine a messy dataset (spreadsheet with errors)
3. Identify at least 5 different types of errors
4. Propose a specific cleaning action for each error

Example: Messy Customer Dataset

CustomerID	Name	Age	Email	Salary	JoinDate
1	Alice	25	alice@mail.com	50000	2023-01-15
2	Bob	-5	bob@mail.com	75000	2023-02-20
3	Alice	25	alice@mail.com	50000	2023-01-15
4	Charlie	30	charlie@mail	999999	2023/03/10
5	Diana	28	diana@mail.com	52000	2023-04-05
6	Eve	18	eve@mail.com	45000	01-05-2023

Identifying Errors

Error Type	Example	Proposed Cleaning Action
Duplicate	Customer 1 and 3 are identical	Remove duplicate row
Invalid Age	Age -5 is impossible	Remove or impute (maybe set to null)
Outlier	Salary 999999 is extremely high	Remove, cap, or investigate
Inconsistent Date	2023/03/10 vs 2023-04-05	Standardise to YYYY-MM-DD
Missing Data	No email for Charlie	Flag as missing or impute
Invalid Format	Different date formats	Standardise all dates

7. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction & Hook	5 min	"Gourmet Chef" analogy; introduce GIGO principle
Data Cleaning Lecture	20 min	Importance of data cleaning; techniques for missing data, duplicates, outliers, formatting; normalisation/standardisation
Practical Activity	15 min	"Clean the Sheet" — students identify errors and propose cleaning actions
Feature Selection Lecture	15 min	Role of feature selection; filter, wrapper, embedded methods
Consolidation	5 min	Recap; preview next lesson

Discussion Questions

Question	Purpose
"Why is data cleaning more important than the model itself?"	Understand GIGO
"When would you use deletion instead of imputation for missing data?"	Understand trade-offs
"What is the difference between normalisation and standardisation?"	Differentiate scaling techniques
"Why is feature selection important?"	Understand the purpose
"What is the difference between filter and wrapper methods?"	Compare feature selection strategies

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
"Clean the Sheet" Activity	Assess ability to identify errors and propose cleaning actions
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the GIGO principle?
2. What is the difference between normalisation and standardisation?
3. Give one method for handling missing data.
4. What is the purpose of feature selection?
5. Name the three feature selection strategies.

9. Differentiation

Level	Strategy
Support	Use analogies; provide clean/ dirty dataset examples; work through the "Clean the Sheet" activity together
Challenge	Research advanced imputation techniques; explore feature selection in real-world datasets

10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.2.1 & A4.2.2 with examples)
- **Worksheet:** "Clean the Sheet" activity
- **Worksheet Answers**
- **Quizlet** (A4.2.1/A4.2.2 vocabulary flashcards)

11. Summary: Key Takeaways

Concept	Key Point
GIGO	Poor data quality leads to poor model performance
Missing Data	Handle via deletion or imputation
Duplicates	Remove duplicate records
Outliers	Remove, cap, or transform
Inconsistent Formats	Standardise to a common format
Normalisation	Scale data to a range (0 to 1)
Standardisation	Scale data to mean 0, std 1
Filter Method	Feature selection BEFORE the model
Wrapper Method	Feature selection USING the model
Embedded Method	Feature selection DURING model training

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KEY REMINDERS

✔ Data quality is more important than model quality

✔ GIGO: Garbage In, Garbage Out

✔ Missing data = deletion or imputation

✔ Outliers = remove, cap, or transform

✔ Normalisation = 0 to 1 range

✔ Standardisation = mean 0, std 1

✔ Filter = before model

✔ Wrapper = using model

✔ Embedded = during model

| *"Good data is the foundation of good machine learning—clean your data, choose your features wisely, and your models will perform better."*